DPT Exercise 11

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Start: 12:15

DPT Exercises

Exercise sheet 11 — Niki Kilbertus

Exercise 1 (hypothesis test for a mean)**.** A factory produces light bulbs, and the historical average lifetime is $\mu_0 = 1200$ hours with a known standard deviation of $\sigma = 150$ hours. A new manufacturing process is introduced, and the manager wants to test if the average lifetime has changed. A sample of $n = 36$ light bulbs produced by the new process has a sample mean lifetime of $\bar{x}_n = 1250$ hours. Assume the lifetimes are normally distributed.

(i) State the null hypothesis H_0 and the alternative hypothesis H_1 for this test.

(ii) Calculate the value of the appropriate test statistic.

(iii) Would you reject the null at a significance level of $\alpha = 0.05$? What about at $\alpha = 0.001$? What does that mean for the confidence you should have in the rejection?

Exercise 2 (Neyman-Pearson test for exponential distribution)**.** Suppose we observe a single sample X from an exponential distribution $\mathrm{Exp}(\lambda)$. We want to test the simple null hypothesis $H_0 : \{\lambda_0\}$ against the simple alternative hypothesis $H_1 : \{\lambda_1\}$, where $\lambda_1 > \lambda_0 > 0$.

(i) Write down the likelihood ratio $\Lambda(x) = \frac{L(\lambda_1|x)}{L(\lambda_0|x)}$.

(ii) According to the Neyman-Pearson lemma, the most powerful test of size α rejects H_0 if $\Lambda(x) > k'$ for some constant k'. Show that this rejection region is equivalent to $x < c$ or $x > c$ for some constant c. Determine which form is correct for $\lambda_1 > \lambda_0$.

(iii) For a significance level α, determine the explicit form of the critical value c for the rejection region.

(iv) What is the power of this test?

Exercise 3 (one-sample t-test for a mean)**.** A brewery claims that its bottles contain exactly 330 ml of beer. A skeptical student measures the contents of $n = 10$ randomly selected bottles and finds a sample mean of $\bar{x}_{10} = 328.5$ ml and a sample variance of $s_{10}^2 = 4$ ml^2. Assuming the amount of beer in a bottle is normally distributed.

(i) State the null and alternative hypotheses for testing if the true mean amount of beer is different from 330 ml.

(ii) What is the appropriate test for this situation? State the test statistic and its distribution under the null hypothesis.

(iii) Perform the test at a significance level of $\alpha = 0.05$. Should the student reject the brewery's claim? [Hint: You may use that the 0.975-quantile of a t-distribution with 9 degrees of freedom is $t_{9,0.025} \approx 2.262$.]

Exercise 4 (type II error and power of a z-test)**.** Suppose you are testing the mean of a normal population $\mathcal{N}(\mu, \sigma^2)$ with known variance $\sigma^2 = 9$. You collect $n = 25$ samples. You want to test the null hypothesis $H_0 : \mu \leq 10$ against the alternative $H_1 : \mu > 10$ at a significance level of $\alpha = 0.01$.

(i) Determine the rejection region for this test. You may use that the 0.99-quantile of a standard normal distribution is $z_{0.01} \approx 2.33$.

(ii) Calculate the probability of a type II error, β, if the true mean is actually $\mu_a = 11.5$.

(iii) What is the power of the test against the alternative $\mu_a = 11.5$?

Exercise 1 (hypothesis test for a mean). A factory produces light bulbs, and the historical average lifetime is $\mu_0 = 1200$ hours with a known standard deviation of $\sigma = 150$ hours. A new manufacturing process is introduced, and the manager wants to test if the average lifetime has changed. A sample of $n = 36$ light bulbs produced by the new process has a sample mean lifetime of $\bar{x}_n = 1250$ hours. Assume the lifetimes are normally distributed.

H_0: innocent (status-quo) H_1: until proven guilty (claim)

(i) State the null hypothesis H_0 and the alternative hypothesis H_1 for this test.

(ii) Calculate the value of the appropriate test statistic.

z^c: critical value

(iii) Would you reject the null at a significance level of $\alpha = 0.05$? What about at $\alpha = 0.001$? What does that mean for the confidence you should have in the rejection?

(i) $H_0: \mu = \mu_0 = 1200$; $H_1: \mu \neq 1200$ (two-sided test)

($H_0: \mu \leq 1200$; $H_1: \mu > 1200$ ("manager wants to test if avg. lifetime has increased")) •

($H_0: \mu \geq 1200$; $H_1: \mu < 1200$ ("... decreased"))

(iii) RV of interest assumed to be normally distributed
- σ known → z-Test
- σ unknown / sample s_n given → t-Test (heavier tails; converges to ND. as sample size $\to \infty$)

$$z = \frac{\bar{X}_n - \mu_0}{\sigma/\sqrt{n}} = \frac{1250 - 1200}{150/\sqrt{36}} = \frac{50}{150} \cdot 6 = 2 : z_{obs}$$

rejection criterion: $|z| > z^c_\alpha$

Exercise 1 (hypothesis test for a mean). A factory produces light bulbs, and the historical average lifetime is $\mu_0 = 1200$ hours with a known standard deviation of $\sigma = 150$ hours. A new manufacturing process is introduced, and the manager wants to test if the average lifetime has changed. A sample of $n = 36$ light bulbs produced by the new process has a sample mean lifetime of $\bar{x}_n = 1250$ hours. Assume the lifetimes are normally distributed.

H_0: innocent (status quo) H_1: until proven guilty (claim)

(i) State the null hypothesis H_0 and the alternative hypothesis H_1 for this test.

(ii) Calculate the value of the appropriate test statistic.

(iii) Would you reject the null at a significance level of $\alpha = 0.05$? What about at $\alpha = 0.001$? What does that mean for the confidence you should have in the rejection?

z^c: critical value

(i) $H_0: \mu = \mu_0 = 1200 \; ; \; H_1: \mu \neq 1200$ (two-sided test)

(ii) RV of interest assumed to be normally distributed, σ known $\rightarrow$ z-Test

$$z = \frac{\bar{X}_n - \mu_0}{\sigma/\sqrt{n}} = \frac{1250 - 1200}{150/\sqrt{36}} = \frac{50}{150} \cdot 6 = 2 : z_{obs}$$

rejection criterion: $|z| > z^c_\alpha = z_{1-\alpha/2}$

$\alpha = 0.05$: $z^c = z_{0.975} = 1.96$: $|z| = 2 > 1.96 \Rightarrow$ reject H_0

$\alpha = 0.001$: $z^c = z_{0.9995} = 3.29$: $|z| = 2 \not> 3.29 \Rightarrow$ not enough evidence to reject H_0

rejection at $\alpha = 0.01$

$\rightarrow$ rejection at all $\alpha' > 0.01$

$z_{0.975} = 1.96$

$z_{0.9995} = 3.29$

Type I: H_0 rejected | H_0

Type II: H_0 not rejected | H_1

$\alpha = P(\text{Type I})$

$\beta: P(\text{Type II})$

Exercise 2 (Neyman-Pearson test for exponential distribution)**.** Suppose we observe a single sample X from an exponential distribution $\mathrm{Exp}(\lambda)$. We want to test the simple null hypothesis $H_0 : \{\lambda_0\}$ against the simple alternative hypothesis $H_1 : \{\lambda_1\}$, where $\lambda_1 > \lambda_0 > 0$.

- $X_{\mathbb{R}_{>0}}(x)$ / assume $x > 1$ C

(i) Write down the likelihood ratio $\Lambda(x) = \frac{L(\lambda_1|x)}{L(\lambda_0|x)}$.

for arbitrary n: $L(\lambda_1 | \vec{x}) = P(X_1 = x_1, \dots, X_n = x_n | \lambda_1) \overset{\text{IID}}{=} \prod_{j=1}^{n} P_{X \sim \mathrm{Exp}(\lambda_1)}(x_j)$

$n = 1$: $L(\lambda_1 | x) = P_{X \sim \mathrm{Exp}(\lambda_1)}(x)$

$$\Rightarrow \quad \Lambda(x) = \frac{\lambda_1 \cdot e^{-\lambda_1 x}}{\lambda_0 \cdot e^{-\lambda_0 x}} = \frac{\lambda_1}{\lambda_0} \cdot e^{(\lambda_0 - \lambda_1) \cdot x}$$

Exercise 2 (Neyman-Pearson test for exponential distribution)**.** Suppose we observe a single sample X from an exponential distribution $\mathrm{Exp}(\lambda)$. We want to test the simple null hypothesis $H_0 : \{\lambda_0\}$ against the simple alternative hypothesis $H_1 : \{\lambda_1\}$, where $\lambda_1 > \lambda_0 > 0$.

(ii) According to the Neyman-Pearson lemma, the most powerful test of size α rejects H_0 if $\Lambda(x) > k'$ for some constant k'. Show that this rejection region is equivalent to $x < c$ or $x > c$ for some constant c. Determine which form is correct for $\lambda_1 > \lambda_0$.

$$\Lambda(x) = \frac{\lambda_1}{\lambda_0} \cdot e^{(\lambda_0 - \lambda_1)\cdot x} > k' \qquad \lambda_1, \lambda_0, k' > 0$$

$$\Leftrightarrow \quad e^{(\lambda_0 - \lambda_1) x} > \underbrace{\frac{\lambda_0}{\lambda_1} \cdot k'}_{=: k''}$$

$$\Leftrightarrow \quad \underbrace{(\lambda_0 - \lambda_1)}_{< 0} x > \ln(k'')$$

$$\Leftrightarrow \quad x < \frac{\ln(k'')}{\lambda_0 - \lambda_1} =: c$$

Theorem 3.40 (Neyman-Pearson Lemma, 1932). *Consider a binary statistical model $(\Omega, \mathcal{A}, (P_{\theta_0}, P_{\theta_1}))$ in which we test the simple null hypothesis $H_0 : \{\theta_0\}$ against the simple alternative $H_1 : \{\theta_1\}$. Let $L(\theta \mid x)$ be the likelihood function. For a given significance level $\alpha \in (0, 1)$, the most powerful test is the* likelihood ratio test (lrt)*, which rejects H_0 if and only if*

$$\frac{L(\theta_1 \mid x)}{L(\theta_0 \mid x)} > k \,,$$

where the constant k is chosen such that the test has size α (implying that such a test exists).

Observation:

$\Lambda(x) > 0 \,, \; x > 0 \quad \Rightarrow \quad k', c > 0$

Exercise 2 (Neyman-Pearson test for exponential distribution). Suppose we observe a single sample X from an exponential distribution $\mathrm{Exp}(\lambda)$. We want to test the simple null hypothesis $H_0 : \{\lambda_0\}$ against the simple alternative hypothesis $H_1 : \{\lambda_1\}$, where $\lambda_1 > \lambda_0 > 0$.

(iii) For a significance level α, determine the explicit form of the critical value c for the rejection region.

$X \sim Exp(\lambda)$ (under $H_0: \lambda = \lambda_0$)

$\Rightarrow p_X(x) = P(X = x) = \lambda \cdot e^{-\lambda \cdot x} \cdot \chi_{\mathbb{R}_{\geq 0}}(x)$

$F_X(x) = P(X \leq x) = (1 - e^{-\lambda x}) \cdot \chi_{\mathbb{R}_{\geq 0}}(x)$

(Here assuming $x \geq 0$)

$\alpha = P(\text{Type I error}) = P(H_0 \text{ rejected} \mid H_0) = P(X < c \mid \lambda = \lambda_0)$

rejection criterion

$= F_{X \sim Exp(\lambda_0)}(c) = 1 - e^{-\lambda_0 x} \Big|_{x=c}$

$= 1 - e^{-\lambda_0 c} \iff e^{-\lambda_0 c} = 1 - \alpha \iff -\lambda_0 c = \ln(1-\alpha) \; ; \lambda_0 > 0$

(solving for c) $\iff c = -\frac{\ln(1-\alpha)}{\lambda_0}$

$\Rightarrow H_0$ rejected iff $x < -\frac{\ln(1-\alpha)}{\lambda_0}$

Exercise 2 (Neyman-Pearson test for exponential distribution)**.** Suppose we observe a single sample X from an exponential distribution $\mathrm{Exp}(\lambda)$. We want to test the simple null hypothesis $H_0 : \{\lambda_0\}$ against the simple alternative hypothesis $H_1 : \{\lambda_1\}$, where $\lambda_1 > \lambda_0 > 0$.

(iv) What is the power of this test?

The *power function* of a test is

$$G_\varphi : \Theta \to [0,1],\ \theta \mapsto \mathbb{E}_\theta[\varphi]\ .$$

For $\theta \in \Theta_1$, we call $G_\varphi(\theta)$ the *power of φ against (the alternative) θ*. The power is the probability with which the alternative $\theta \in \Theta_1$ is recognized, when it is true. The probability of a type II error at $\theta \in \Theta_1$ is then $\beta(\theta) := 1 - G_\varphi(\theta)$.

Prob. of Type II error

Generally: $G_\varphi(\lambda_1) = 1 - \beta(\lambda_1)$

"power of a test" $= P($"correctly rejecting H_0"$) = P(H_0 \text{ rejected} \mid H_1) = P(X < c \mid \lambda_1)$

$= F_{X \sim \mathrm{Exp}(\lambda_1)}(c) = 1 - e^{-\lambda_1 c}$ $\Big|$ $c = \dfrac{-\ln(1-\alpha)}{\lambda_0}$

$= 1 - \exp\left(-\lambda_1 \cdot \left(-\frac{1}{\lambda_0}\right) \cdot \ln(1-\alpha)\right) = 1 - (1-\alpha)^{\lambda_1/\lambda_0}$

$= 1 - P(X \geq c \mid \lambda_1)$

$= 1 - P(H_0 \text{ not rejected} \mid H_1)$

$= 1 - P($"Type II error"$)$

rel. to $\beta(\lambda_1)$

Exercise 3 (one-sample t-test for a mean). A brewery claims that its bottles contain exactly 330 ml of beer. A skeptical student measures the contents of $n = 10$ randomly selected bottles and finds a sample mean of $\bar{x}_{10} = 328.5$ ml and a sample variance of $s_{10}^2 = 4\ \text{ml}^2$. Assuming the amount of beer in a bottle is normally distributed. ⇒ t-Test

(i) State the null and alternative hypotheses for testing if the true mean amount of beer is different from 330 ml.

(ii) What is the appropriate test for this situation? State the test statistic and its distribution under the null hypothesis.

(i) $H_0: \mu = \overbrace{330}^{=:\mu_0}$, $H_1: \mu \neq 330$ (two-sided)

(ii) t-Test: $T = \dfrac{\bar{X}_n - \mu_0}{\sqrt{S_n^2 \cdot n}} = \dfrac{328.5 - 330}{\sqrt{4 \cdot 10}} \approx -2.373$

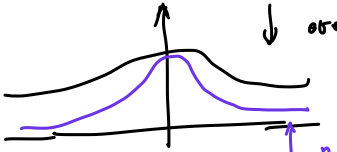
rejection criterion: $|T| > t^c = t_{10-1;\,0.975} \approx 2.262$

$|T| = 2.373 > 2.262 \Rightarrow$ reject H_0

Student's t-distribution

$p_X(\,\cdot \mid df = n-1)$

sample size → df small.
heavier tails
↓ overall „flatter"



$n \to \infty$
$\xrightarrow{d} N(\mu, \sigma^2)$

Exercise 3 (one-sample t-test for a mean)**.** A brewery claims that its bottles contain exactly 330 ml of beer. A skeptical student measures the contents of $n = 10$ randomly selected bottles and finds a sample mean of $\bar{x}_{10} = 328.5$ ml and a sample variance of $s_{10}^2 = 4$ ml^2. Assuming the amount of beer in a bottle is normally distributed.

(iii) Perform the test at a significance level of $\alpha = 0.05$. Should the student reject the brewery's claim? [Hint: You may use that the 0.975-quantile of a t-distribution with 9 degrees of freedom is $t_{9,0.025} \approx 2.262$.]

Exercise 4 (type II error and power of a z-test). Suppose you are testing the mean of a normal population $\mathcal{N}(\mu, \sigma^2)$ with known variance $\sigma^2 = 9$. You collect $n = 25$ samples. You want to test the null hypothesis $H_0 : \mu \leq 10$ against the alternative $H_1 : \mu > 10$ at a significance level of $\alpha = 0.01$.

⇒ z-Test

(i) Determine the rejection region for this test. You may use that the 0.99-quantile of a standard normal distribution is $z_{0.01} \approx 2.33$.

$H_0: \mu \leq 10$, $H_1: \mu > 10$

$$T = \frac{\bar{X}_n - \mu_0}{\sigma/\sqrt{n}} = \frac{\bar{X}_n - 10}{\sqrt{9}/\sqrt{25}} = \frac{\bar{X}_n - 10}{0.6}$$

Rejection criterion: $T > t^c = t_{0.99} = 2.33$

$$T > t^c \iff \frac{\bar{X}_n - 10}{0.6} > 2.33$$

$$\iff \bar{X}_n > 10 + 0.6 \cdot 2.33 = 11.398$$

rejection region: $\bar{X}_n \in (11.398, \infty) =: I^c$

Exercise 4 (type II error and power of a z-test)**.** Suppose you are testing the mean of a normal population $\mathcal{N}(\mu, \sigma^2)$ with known variance $\sigma^2 = 9$. You collect $n = 25$ samples. You want to test the null hypothesis $H_0 : \mu \leq 10$ against the alternative $H_1 : \mu > 10$ at a significance level of $\alpha = 0.01$.

(ii) Calculate the probability of a type II error, β, if the true mean is actually $\mu_a = 11.5$.

replace with $\mu = \mu_a = 11.5$

$$P(\text{"type II error"}) = P(H_0 \text{ not rejected} \mid H_1) = P(\bar{X}_n \notin I^c \mid \mu > 10)$$

$$= P(\bar{X}_n \leq 11.348 \mid \mu = 11.5)$$

$$X \sim N(11.5, 9)$$

$$\Rightarrow \bar{X}_n \sim N\left(11.5, \tfrac{9}{25}\right)$$

$$= P\left(\frac{\bar{X}_n - 11.5}{\sqrt{9/25}} \leq \frac{11.348 - 11.5}{\sqrt{9/25}}\right) = \Phi(-0.17) \approx 0.4325$$

$\frac{\bar{X}_n - 11.5}{\sqrt{9/25}} \sim N(0,1)$

$\frac{11.348 - 11.5}{\sqrt{9/25}} = -0.17$

Exercise 4 (type II error and power of a z-test)**.** Suppose you are testing the mean of a normal population $\mathcal{N}(\mu, \sigma^2)$ with known variance $\sigma^2 = 9$. You collect $n = 25$ samples. You want to test the null hypothesis $H_0 : \mu \leq 10$ against the alternative $H_1 : \mu > 10$ at a significance level of $\alpha = 0.01$.

(iii) What is the power of the test against the alternative $\mu_a = 11.5$?

$G_\varphi(\lambda_1) = 1 - \beta(\lambda_1) = 1 - 0.4325 = 0.5675$

Prob of Type II error under λ_1

Overview 3.33 (hypothesis testing motivation). A *statistical hypothesis* is a claim about the parameters of a statistical model. In hypothesis testing, we typically consider two competing, mutually exclusive hypotheses.

- The *null hypothesis*, denoted H_0, is a statement of no effect or no difference, often representing a default or status quo position. It consists of a subset $\Theta_0 \subseteq \Theta$ of the parameter space.
- The *alternative hypothesis*, denoted H_1 (or H_A), is a statement that contradicts the null hypothesis, representing the claim we wish to find evidence for. It specifies a subset $\Theta_1 \subseteq \Theta$ such that $\Theta_0 \cap \Theta_1 = \emptyset$.

Definition 3.34 (building blocks & properties of hypothesis tests). Let $(\Omega, \mathcal{A}, P_\Theta)$ be a statistical model and let $\Theta = \Theta_0 \mathbin{\mathring{\cup}} \Theta_1$ be a partition of the parameter space.

(i) A hypothesis is *simple* if $|\Theta_0| = 1$ (or $|\Theta_1| = 1$) and we then denote the single element by θ_0 (or θ_1). Otherwise ($|\Theta_{0/1}| > 1$), it is called *composite*. Often, there is also a special $\theta_{0/1} \in \Theta_{0/1}$ for composite statements, e.g., $\Theta_0 = \{\theta \in \Theta \mid \theta > \theta_0\}$.

(ii) For $X \sim P_\Theta$, the *null hypothesis H_0 is valid* if $(\Omega, \mathcal{A}, P_{\Theta_0})$ is well-specified. Otherwise, the *alternative H_1 is valid*.

(iii) A *statistical test* is a measurable map

$$\varphi : \Omega^n \to [0, 1] \,,$$

which is interpreted as a decision rule for whether to reject the null or not. If $\varphi(\mathcal{X}) \subseteq \{0, 1\}$, we call the test *non-randomized*, otherwise we call it *randomized*. We focus primarily on non-randomized tests.

Definition 3.34 (building blocks & properties of hypothesis tests). Let $(\Omega, \mathcal{A}, P_\Theta)$ be a statistical model and let $\Theta = \Theta_0 \mathbin{\mathring{\cup}} \Theta_1$ be a partition of the parameter space.

(iv) For non-randomized tests, by convention we interpret $\varphi(x) = 1$ as rejecting the null. The set $\varphi^{-1}(\{1\}) = \{x \in \Omega^n \mid \varphi(x) = 1\}$ is called *rejection region*.

- Rejecting the null hypothesis H_0 when it is actually true, i.e., $\varphi(x) = 1$ albeit P_{Θ_0} well-specified, is called *type I error*.
- Failing to reject the null hypothesis H_0 when it is actually false, i.e., $\varphi(x) = 0$ albeit P_{Θ_1} well-specified, a *type II error*.

	decision	
true state	fail to reject H_0	reject H_0
H_0 is true	correct decision	type I error
H_1 is true	type II error	correct decision

(v) The *significance level* or *size* of a test φ is the maximum probability of a type I error

$$\alpha := \sup_{\theta \in \Theta_0} \mathbb{E}_\theta[\varphi] \,.$$

A test φ is called *test at (significance) level* α for $\alpha \in [0, 1]$ if $\sup_{\theta \in \Theta_0} \mathbb{E}_\theta[\varphi] \leq \alpha$.

Theorem 3.40 (Neyman-Pearson Lemma, 1932). *Consider a binary statistical model* $(\Omega, \mathcal{A}, (P_{\theta_0}, P_{\theta_1}))$ *in which we test the simple null hypothesis* $H_0 : \{\theta_0\}$ *against the simple alternative* $H_1 : \{\theta_1\}$. *Let* $L(\theta \mid x)$ *be the likelihood function. For a given significance level* $\alpha \in (0, 1)$, *the most powerful test is the* likelihood ratio test (lrt), *which rejects* H_0 *if and only if*

$$\frac{L(\theta_1 \mid x)}{L(\theta_0 \mid x)} > k \,,$$

where the constant k *is chosen such that the test has size* α *(implying that such a test exists).*

Definition 3.43 (one/two-sided tests). Let $\Theta \subset \mathbb{R}$.

- A *one-sided* (or *one-tailed*) test is used for directional hypotheses, such as $H_1 : \theta > \theta_0$ or $H_1 : \theta < \theta_0$. The rejection region is entirely in one tail of the distribution of the test statistic.
- A *two-sided* (or *two-tailed*) test is used for non-directional hypotheses, such as $H_1 : \theta \neq \theta_0$. The rejection region is split between the two tails of the distribution.

name	H_0	statistic	null distr.
single mean μ			
z-test (σ^2 known)	$\mu = \mu_0$	$Z = \frac{\bar{X}_n - \mu_0}{\sigma/\sqrt{n}}$	$\mathcal{N}(0,1)$
t-test (σ^2 unknown)	$\mu = \mu_0$	$T = \frac{\bar{X}_n - \mu_0}{S_n/\sqrt{n}}$	t_{n-1}
single variance σ^2			
χ^2-test (μ unknown)	$\sigma^2 = \sigma_0^2$	$C = \frac{(n-1)S_n^2}{\sigma_0^2}$	χ^2_{n-1}
comparing means μ_X, μ_Y			
two-sample t-test (equal variances)	$\mu_X = \mu_Y$	$T = \frac{\bar{X} - \bar{Y}}{S_p\sqrt{\frac{1}{n_X} + \frac{1}{n_Y}}}$	$t_{n_X+n_Y-2}$

Definition 3.48 (p-value). The viewpoint 3.42 and definition 3.43 give rise to the idea of a *p-value*, which is the probability, assuming the null hypothesis H_0 is true, of observing a test statistic at least as extreme as the one computed from the sample data. The direction of "extremeness" is determined by the alternative H_1. For an observed test statistic t_{obs}:

- if $H_1 : \theta > \theta_0$, p-value $= P_{\theta_0}(T(X) \geq t_{\text{obs}})$.
- if $H_1 : \theta < \theta_0$, p-value $= P_{\theta_0}(T(X) \leq t_{\text{obs}})$.
- if $H_1 : \theta \neq \theta_0$, p-value $= P_{\theta_0}(T(X)$as or more extreme than $t_{\text{obs}})$.

The test is then $\varphi(x) = 1$ if the p-value is less than or equal to the chosen significance level $\alpha \in [0,1]$.

Definition 3.54 (confidence interval). A $100(1-\alpha)\%$ *confidence interval (ci)* for a parameter of interest $\gamma(\theta) \in \mathbb{R}$ is an interval with random endpoints $(L(X), U(X))$, computed from the sample $X = (X_1, \ldots, X_n)$, such that

$$P_\theta(L(X) \leq g(\theta) \leq U(X)) \geq 1 - \alpha \quad \text{for all } \theta \in \Theta\,.$$

The value $1-\alpha$ is the *confidence level*. The key is that the interval is random, while the parameter is treated as a fixed, unknown constant.

z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
+0	.50000	.50399	.50798	.51197	.51595	.51994	.52392	.52790	.53188	.53586
+0.1	.53983	.54380	.54776	.55172	.55567	.55966	.56360	.56749	.57142	.57535
+0.2	.57926	.58317	.58706	.59095	.59483	.59871	.60257	.60642	.61026	.61409
+0.3	.61791	.62172	.62552	.62930	.63307	.63683	.64058	.64431	.64803	.65173
+0.4	.65542	.65910	.66276	.66640	.67003	.67364	.67724	.68082	.68439	.68793
+0.5	.69146	.69497	.69847	.70194	.70540	.70884	.71226	.71566	.71904	.72240
+0.6	.72575	.72907	.73237	.73565	.73891	.74215	.74537	.74857	.75175	.75490
+0.7	.75804	.76115	.76424	.76730	.77035	.77337	.77637	.77935	.78230	.78524
+0.8	.78814	.79103	.79389	.79673	.79955	.80234	.80511	.80785	.81057	.81327
+0.9	.81594	.81859	.82121	.82381	.82639	.82894	.83147	.83398	.83646	.83891
+1	.84134	.84375	.84614	.84849	.85083	.85314	.85543	.85769	.85993	.86214
+1.1	.86433	.86650	.86864	.87076	.87286	.87493	.87698	.87900	.88100	.88298
+1.2	.88493	.88686	.88877	.89065	.89251	.89435	.89617	.89796	.89973	.90147
+1.3	.90320	.90490	.90658	.90824	.90988	.91149	.91308	.91466	.91621	.91774
+1.4	.91924	.92073	.92220	.92364	.92507	.92647	.92785	.92922	.93056	.93189
+1.5	.93319	.93448	.93574	.93699	.93822	.93943	.94062	.94179	.94295	.94408
+1.6	.94520	.94630	.94738	.94845	.94950	.95053	.95154	.95254	.95352	.95449
+1.7	.95543	.95637	.95728	.95818	.95907	.95994	.96080	.96164	.96246	.96327
+1.8	.96407	.96485	.96562	.96638	.96712	.96784	.96856	.96926	.96995	.97062
+1.9	.97128	.97193	.97257	.97320	.97381	.97441	.97500	.97558	.97615	.97670
+2	.97725	.97778	.97831	.97882	.97932	.97982	.98030	.98077	.98124	.98169
+2.1	.98214	.98257	.98300	.98341	.98382	.98422	.98461	.98500	.98537	.98574
+2.2	.98610	.98645	.98679	.98713	.98745	.98778	.98809	.98840	.98870	.98899
+2.3	.98928	.98956	.98983	.99010	.99036	.99061	.99086	.99111	.99134	.99158
+2.4	.99180	.99202	.99224	.99245	.99266	.99286	.99305	.99324	.99343	.99361
+2.5	.99379	.99396	.99413	.99430	.99446	.99461	.99477	.99492	.99506	.99520
+2.6	.99534	.99547	.99560	.99573	.99585	.99598	.99609	.99621	.99632	.99643
+2.7	.99653	.99664	.99674	.99683	.99693	.99702	.99711	.99720	.99728	.99736
+2.8	.99744	.99752	.99760	.99767	.99774	.99781	.99788	.99795	.99801	.99807
+2.9	.99813	.99819	.99825	.99831	.99836	.99841	.99846	.99851	.99856	.99861
+3	.99865	.99869	.99874	.99878	.99882	.99886	.99889	.99893	.99896	.99900
+3.1	.99903	.99906	.99910	.99913	.99916	.99918	.99921	.99924	.99926	.99929
+3.2	.99931	.99934	.99936	.99938	.99940	.99942	.99944	.99946	.99948	.99950
+3.3	.99952	.99953	.99955	.99957	.99958	.99960	.99961	.99962	.99964	.99965
+3.4	.99966	.99968	.99969	.99970	.99971	.99972	.99973	.99974	.99975	.99976
+3.5	.99977	.99978	.99978	.99979	.99980	.99981	.99981	.99982	.99983	.99983
+3.6	.99984	.99985	.99985	.99986	.99986	.99987	.99987	.99988	.99988	.99989
+3.7	.99989	.99990	.99990	.99990	.99991	.99991	.99992	.99992	.99992	.99992
+3.8	.99993	.99993	.99993	.99994	.99994	.99994	.99994	.99995	.99995	.99995
+3.9	.99995	.99995	.99996	.99996	.99996	.99996	.99996	.99996	.99997	.99997
+4	.99997	.99997	.99997	.99997	.99997	.99997	.99998	.99998	.99998	.99998

$z_{0.975} = 1.96$

$z_{0.9995} = 3.29$

$z_{0.99} \approx 2.03$